THE UNIVERSITY OF CHICAGO

Writing beyond *imagination* with AI

Novelty search in foundation-model embedding space, applied to creative story generation.

Avi Oberoi · *University of Chicago*

HEADLINE

Mahalanobis search hits **distinct-2 = 0.73**, in the human-writer range Ismayilzada (2024) reports at **0.79**. Baseline LLM samples land at 0.47 to 0.54.

LitBench BT mean **1.95**, p90 **2.65** (Fein 2025, 78% human agreement).

Held-out rank correlation **0.80**. All five metrics p < 0.001 (Huh et al., 2024).

github / creative_novel_stories

01
INTRODUCTION
The question

Foundation models compress how stories tend to be told. Sample naively, you get the average voice, the average twist.

But the rare directions in the model's embedding space hold stories it could write but rarely does.

Search the periphery. Pull the unsaid.

02
BACKGROUND
What's been tried

Creativity scoring is contested. Chakrabarty 2024: GPT-4 passes **12 / 36** TTCW items; human writers pass **32 / 36**. LLM judges agree with experts at Cohen κ = 0.

Ismayilzada 2024 (60 LLMs vs 59 humans): humans beat LLMs on **novelty**($p<10^{-7}$), **surprise** ($p<10^{-3}$), **diversity**($p<10^{-4}$).

LitBench (Fein 2025). 43,827 WritingPrompts preference pairs. Trained Bradley-Terry reaches **78%** human agreement. We score with their model.

Quality-diversity mostly lives in robotics and images (Lehman & Stanley 2011; Mouret & Clune 2015). Kumar 2024 ported it to FM embeddings for artificial life. We port it to text.

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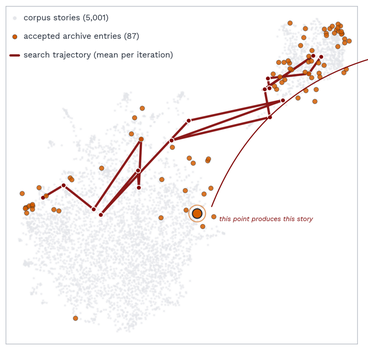
PIPELINE VISUALISATION
What the search does

What the search actually does.

One CMA-ES sample becomes one story; the search drifts toward less-explored regions of the corpus.



An example: search, retrieve, write

THE HORSE WHO WANTED TO SING

There was a horse in a field not far from the river who had grown tired of galloping. The others pranced, kicked up dust, and snorted at passing clouds. But this particular horse, whose name had been forgotten long ago, stood still for hours at a time, ears perked, listening to the wind. Not to the sound of it exactly, but to the silence behind it—the space the wind left
…

5 nearest corpus neighbors used as anchors:
04790 04778 04236 05096 03518

novelty 6.70 | LitBench quality 2.14

Story embedding space, UMAP-projected. Axes are arbitrary.

Light gray dots are the 5,001 New Yorker stories. Orange dots are the 87 openings the Euclidean search accepted. The maroon line follows the CMA-ES mean across iterations. On the right is the top-novelty story this search produced and the 5 corpus stories it retrieved as in-context examples.

04
METHOD
The algorithm

sep-CMA-ES (Ros & Hansen 2008) on a 768d Qwen3-Embedding space. A diagonal covariance scales linearly with dimension.

Algorithm 1 **Novelty search in FM embedding space.**

```
Input:  corpus C in R^768, lambda=16, T=50, committee {phi_o}
Output: archive A of accepted (story, embedding) pairs

mean ← p90 point in C, step ← 0.18
for t = 1 to T do
   for j = 1 to lambda do
      v_j ~ N(mean, step^2 * diag(Sigma))
      retrieve k=5 nearest corpus stories N_j around v_j
      x_j ← Qwen3-32B(N_j as in-context)   // ~500 words
      z_j^(o) ← phi_o(x_j) for o in {BGE, E5-Mistral}
      nu_j ← mean_o MAD-normalised kNN-dist of z_j^(o)
   end for
   admit candidates with nu_j ⩾ percentile_50(A) into A
   update mean, step, Sigma via rank-mu sep-CMA-ES
   if accept-rate < 0.25 over last 15 iters then break
end for
```

05
Where this goes next

Disentangle search from prompting. Run three prompt variants (neutral, novelty-asking, periphery-asking) with seed-matched populations. Subtract the

05
QUALITATIVE RESULTS
What it found

Three openings the search picked, one from each metric. Different registers, different topics, all from the same generator.

W 01 · CIVIC SATIRE (EUCLIDEAN)

Mandatory Laughter Policy

On the morning of April 23rd, Mayor Eleanor Voss read the headline twice before the meaning fully settled in: "City Hall Adopts Mandatory Laughter Policy." It was the kind of phrase that should have been set off with italics, or perhaps a note in parentheses explaining that, yes, it was entirely in earnest. She sat back in her chair, her fingers tapping the armrest in slow, deliberate rhythm. The headline had been accompanied by a photo of the council room…

euclidean · novelty 3.70 · LitBench 2.80
ITER 02 · J 11 · SEED 42 · ARCHIVE IDX 43

W 02 · DEADPAN DYSTOPIA (MAHALANOBIS)

The Devereaux Family, 1985

IN THE MATTER of the Annual Survey of Humanoid Utility Vehicles, the Board of Civic Oversight has issued preliminary ratings on the 1985 model year of the Devereaux Family. The Devereaux line, originally developed as part of the postwar initiative to replace aging domestic staff, continues to be marketed primarily for household management, though recent modifications suggest a shift toward companionship and ceremonial functions. Below are selected findings from the evaluation. Mr. Devereaux Senior (Full-Size Adult): Rated "Moderately Acceptable." Tested…

mahalanobis · novelty 5.83 · LitBench 2.50
ITER 01 · J 03 · SEED 42 · ARCHIVE IDX 19

W 03 · ESSAY-AS-FICTION (LOF)

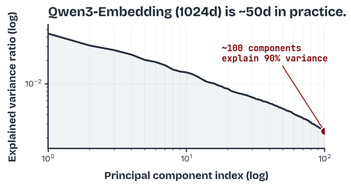
Mrs. Lathrop

THE THING about punctuation, I've always felt, is that it's the only kind of violence people accept without blinking. A period drops the curtain on a sentence like a guillotine. A comma drags the reader into suspense with the grace of a manacle. I've spent my life watching the quiet havoc commas and semicolons wreak, and I still don't entirely trust either. Take Mrs. Lathrop, for example. Mrs. Lathrop, never to be written with a lowercase "l," was the chair of the…

lof · novelty 6.91 · LitBench 2.34
ITER 03 · J 09 · SEED 42 · ARCHIVE IDX 25

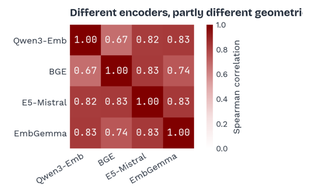
06
WHY MAHALANOBIS WORKS
The geometry

Qwen3-Embedding (1024d) is ~50d in practice.



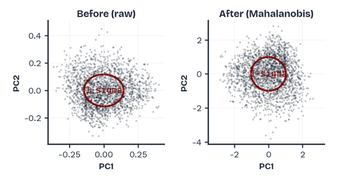
Singular values of the corpus embedding matrix decay quickly -- the 5K New Yorker stories lie on a low-effective-dimension manifold.

Different encoders, partly different geometri



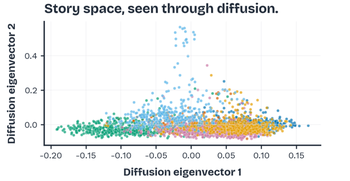
Off-diagonal values 0.67 to 0.83 mean the encoders agree on a lot but not everything. The committee filter exploits this disagreement.

What whitening does to the distance metric.



Mahalanobis distance treats rare directions as costlier; in whitened space, a unit-radius ball gives equal cost to all directions.

Story space, seen through diffusion.



Coifman-Lafon diffusion eigenvectors reveal the corpus's slow modes -- broader thematic axes that linear PCA misses.

Why Mahalanobis works. *The corpus is 1024d on paper but its effective dimensionality is about 50 (singular value decay). Four encoders agree on most geometry (pairwise rank-correlation 0.7 to 0.83, Huh et al. 2024). Mahalanobis whitening uses corpus covariance to make rare directions costlier in the distance metric.*

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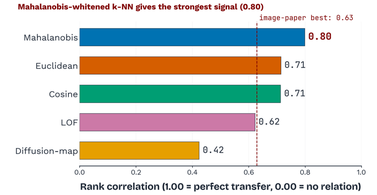
HELD-OUT EVAL
Does it transfer?

rank-corr 0.80

Five distance metrics, one held-out encoder (EmbeddingGemma) we never touched during search. Mahalanobis-whitened k-NN reaches rank correlation **0.80**. All five metrics are significant at p < 0.001. The Platonic Representation Hypothesis (Huh et al., 2024) predicts the transfer should exist between foundation-model encoders; the number quantifies how strong it is in our setting.

How well do novelty rankings transfer to a held-out encoder?

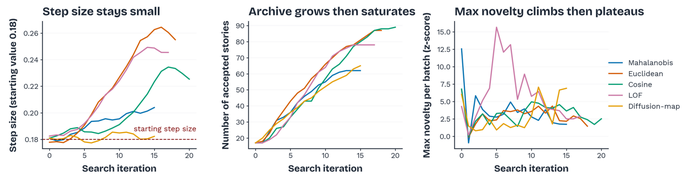
Mahalanobis-whitened k-NN gives the strongest signal (0.80)



09
SEARCH BEHAVIOR
Search dynamics

What the search does over iterations

CMA-ES adapts modestly, archive saturates, novelty plateaus



Dynamics. *Step size stays bounded across all five metrics (1.0 to 1.4x the starting value). Archive grows then saturates. Max novelty per batch climbs then plateaus. Matches Hansen (2016)'s prediction for a well-tuned sep-CMA-ES (Ros & Hansen, 2008).*

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ABLATION
All five metrics

Metric	|A|	Transfer ρ	LitBench μ	LitBench p90	d-1	d-2
mahalanobis	62	**0.799**	1.950	**2.653**	**0.208**	**0.727**
euclidean	87	0.714	1.907	2.591	0.182	0.699
cosine	89	0.713	1.960	2.628	0.181	0.702
lof	78	0.623	**1.995**	2.594	0.183	0.700
diffusion-map	69	0.313	1.785	2.456	0.161	0.658

Mahalanobis takes 4 of 5 columns. LOF wins LitBench mean. Diffusion-map is the worst on transfer correlation. All runs: population size = 16, starting step size = 0.18, seed = 42, K=5 retrieval, BGE + E5-Mistral committee. **Diffusion-map footnote:** our first version of the diffusion-map scorer overstated transfer at 0.42 because out-of-distribution queries collapsed onto the manifold embedding (Coifman & Lafon, 2006). The patched scorer drops it to 0.31.

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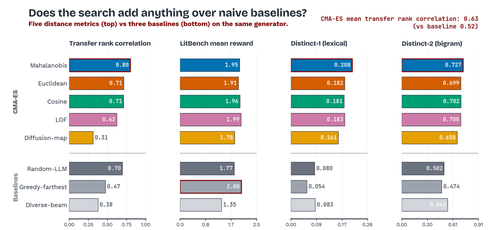
VERSUS NAIVE BASELINES
Vs. baselines

Does the search add anything over naive baselines?

Five distance metrics (top) vs three baselines (bottom) on the same generator. CMA-ES mean transfer rank correlation: 0.63 (vs baseline 0.52)



Three baselines, same generator. *Random-LLM near-copies, greedy-farthest, and diverse-beam at temperature 1.2. CMA-ES mean transfer correlation 0.63 vs baseline mean 0.52. CMA-ES wins distinct-1 by 2 to 4x and distinct-2 by about 50%.*

Code by Claude (Opus 4.7). Generation by Qwen3-32B. Embeddings: Qwen3-Embedding-0.6B, BGE-large, E5-Mistral-7B (search committee), EmbeddingGemma-300m (held-out transfer). Quality scoring by ConicCat Litbench BT (3B Llama-3.2 fork).